LCBI BKM

BKM-TRB-2026-001 Rev 0 Mar 4, 2026

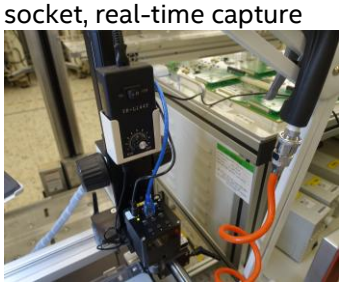
This BKM applies to VCTIM Inspection using the AI-Powered Inspection System

1. Overview

Affected Products and Audience
All products using VCTIM on LCBI Burn In Board (BIB). Intended for production-line operators, quality engineers, and technicians.

Description
This BKM covers the end-to-end procedure for inspecting VCTIMs (Viscous Coupling Thermal Interface Materials) using the AI-powered web inspection application. The system uses a YOLO object-detection model to identify normal and missing VCTIMs in images captured by station or handheld cameras. **Three capture hardware options** are available; each covered in its own section below.

2. Tools & Software Required

S/N	Category	Item / Description
1	Capture Device (pick one)	Fixed Station Camera - for single socket, real-time capture 
1'	Capture Device (pick one)	Handheld Camera - for multiple sockets; requires cable transfer to PC

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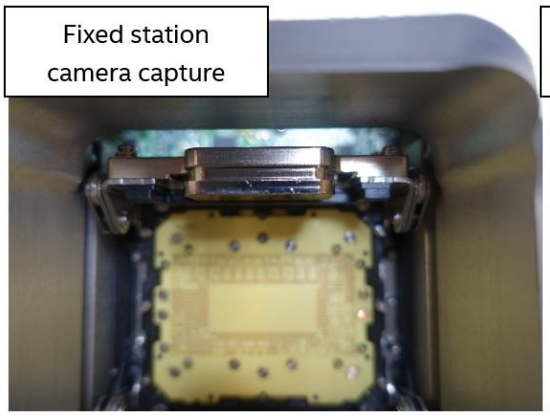
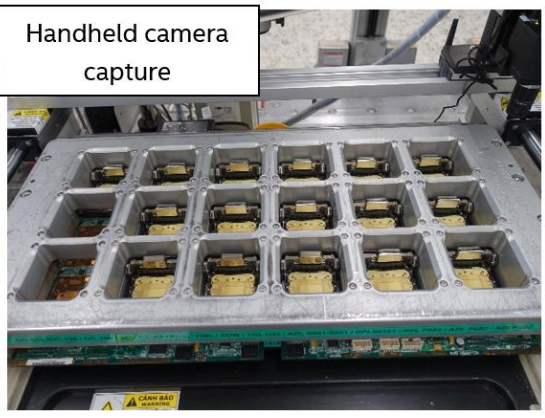
S/N	Category	Item / Description
1"	Capture Device (pick one)	USB Connected Camera - for multiple sockets, real-time capable; no cable transfer needed
		
2	Workstation	PC / laptop with modern web browser (Chrome, Edge, or Firefox)
3	Scanner (optional)	Station barcode scanner - for scanning Unit IDs into the system
4	Lighting (optional)	Station lights - recommended for better image quality (especially multi-socket)

S/N	Software / Resource
1	VCTIM Inspection App, accessible at *http://localhost:8501* (or server IP)
2	Docker (for first-time deployment only) used to run the application container

3. Hardware Setup & Image Capture

Select the appropriate hardware option for your station and follow the corresponding steps below.

PART 1 - Hardware Setup & Image Capture

Option A: Fixed Station Camera

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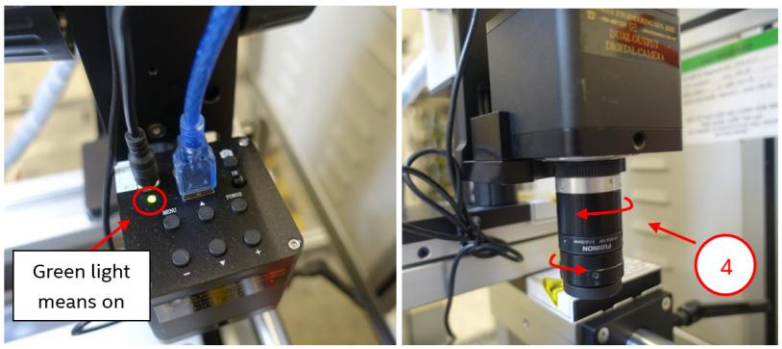
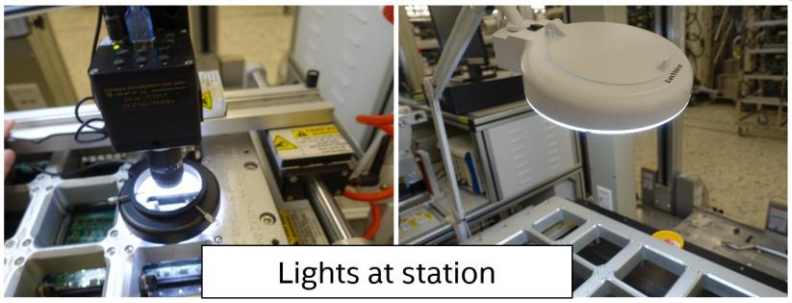
USE CASE: Individual socket capture only. Supports real-time webcam mode within the inspection app.

Step / Description	Action
Step A-1 *Locate the camera*	1. Identify the fixed station camera mounted above the socket test station. 2. Verify the camera is powered on and stable.
Step A-2 *Calibrate for clear image*	3. Look through the camera viewfinder or live feed. 4. Twist the camera lens ring clockwise or counter-clockwise until the VCTIM surface comes into sharp focus. 5. Check that the entire socket face is visible in the frame - no VCTIM should be cut off at the edges.  **TIP:** Use the station lights to reduce shadows and improve detection accuracy. 
Step A-3 *Capture image*	6. Once focused, the camera is ready. 7. Capture will occur in the app via real-time webcam feed (see Part 2, Section A, Step 3).

Option B: Handheld Camera

USE CASE: Multiple socket capture in a single photo. Requires cable transfer to the PC before uploading to the app. Real-time mode is NOT supported.

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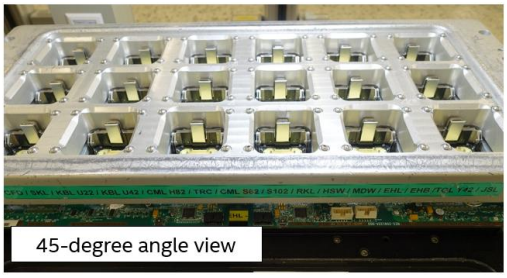
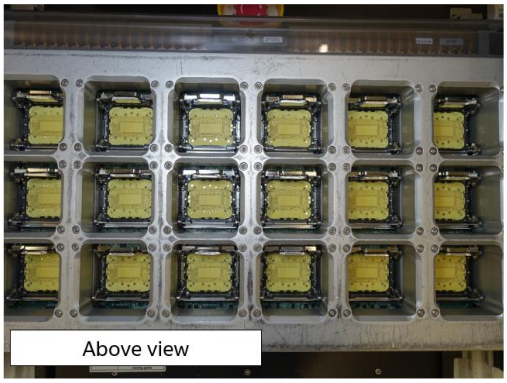
Step / Description	Action
Step B-1 *Retrieve the camera*	1. Open the station equipment cabinet. 2. Retrieve the handheld camera and USB cable stored (indicated by the yellow label). 3. Verify battery charge is sufficient before proceeding.
Step B-2 *Capture the image*	1. Position yourself **above** or at a **45-degree angle** to the socket array so all VCTIMs are visible. 2. Ensure all sockets are clearly visible within the frame - use a bird's-eye view or 45-degree angle. 3. Turn on station lights for better image quality. 4. Take the photo. Confirm the image is sharp and all VCTIMs are in-frame.  45-degree angle view  Above view Bad capture

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Step / Description	Action
Step B-3 *Transfer image to PC*	1. Connect the camera to the workstation PC using the USB data cable stored in the cabinet drawer. 2. Copy the captured image file(s) to a known folder on the PC (ex: Desktop or Downloads). 3. Safely eject the camera after transfer. **NOTE:** The handheld camera does not support real-time capture. Images must be uploaded manually in Part 2.

Option C: USB Connected Camera

USE CASE: Multiple socket capture. Camera stays connected to the PC - no cable transfer needed. Supports real-time webcam mode within the inspection app.

Step / Description	Action
Step C-1 *Connect the camera*	1. Plug the USB camera into an available USB port on the workstation PC. 2. Wait for the OS to recognize the device (typically 5-10 seconds). 3. Verify the camera indicator light is on.
Step C-2 *Position the camera*	1. Mount or position the USB camera above the socket array at a bird's-eye or 45-degree angle so all sockets are visible. 2. Ensure all VCTIMs are within the camera frame. 3. Turn on station lights for better image quality.
Step C-3 *Capture*	1. Images can be captured either through the real-time webcam feed in the app (see Part 2, Section 5a), or by taking a snapshot and uploading it (Section 5b).

4. Inspection System Procedure

Complete Steps 1-8 in the inspection application for every inspection session.

PART 2 - Inspection System Procedure

Step / Description	Action
Step 1 *Open the Inspection App*	1. Open a web browser (Chrome, Edge, or Firefox). 2. Navigate to: *http://localhost:8501* *If the system is hosted on a remote server, use the server IP address instead (ex: http://192.168.x.x:8501).*

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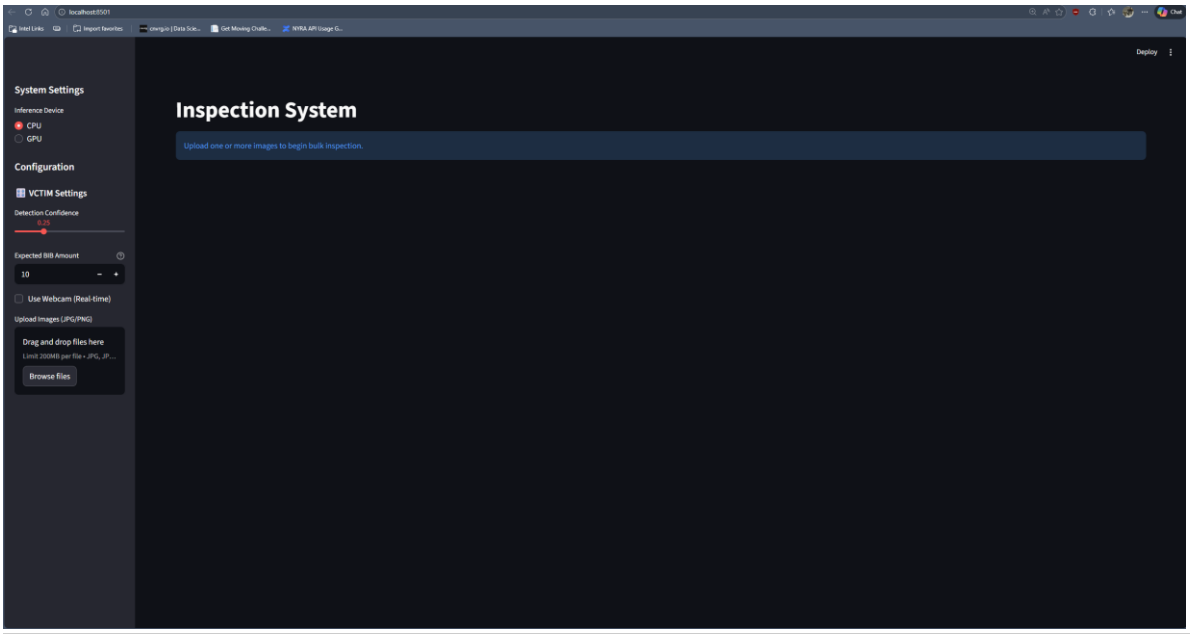
Step / Description	Action
	3. Wait for the page to fully load. You should see the title 'Inspection System' and the sidebar on the left.  **FIRST TIME?** If the app does not load, ensure the Docker container is running. Contact your system administrator for setup. Default command: `*docker-compose up*` (run from the application source folder).
Step 2 *Configure System Settings*	In the sidebar, complete the following settings in order: **1 Inference Device (CPU or GPU)** • Select CPU for standard use. • Select GPU if a compatible NVIDIA GPU is available - this provides faster processing for large batches. • If GPU is unavailable, the system automatically falls back to CPU. **2 Detection Confidence Threshold** • This slider (0.0 - 1.0) sets how confident the AI must be before labelling a detection. • Default value: 0.25. Lower values are more sensitive (may produce more false positives). Higher values are stricter (may miss detections). • Recommended starting value: 0.25. Adjust based on your product and image quality. **3 Expected DUT (Device under Testing) Amount (IMPORTANT)** • Set this number to the total count of VCTIM positions on the board being inspected (including all filled and known-empty positions). • The system uses this number to validate the inspection results. • Consult your product specification sheet for the correct value. **WARNING:** Incorrect DUT Amount will cause false mismatch alerts. Always verify this value before starting an inspection batch.

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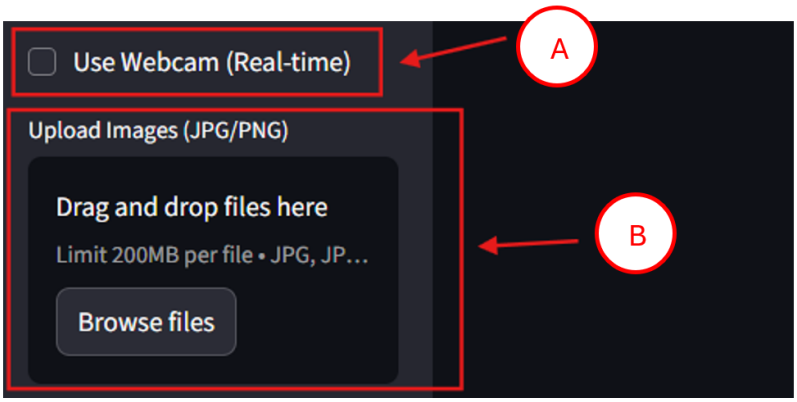
Step / Description	Action
Step 3 *Provide Image(s) to the System*	Choose one of the two input methods below based on your hardware: **Option A: Real-Time Webcam (Fixed Station Camera or USB Camera only)** 1. In the sidebar, tick the checkbox labelled Use Webcam (Real-time). 2. A live camera feed will appear in the main content area. 3. Position the board under the camera. When the image is clear and all VCTIMs are visible, click "Take Photo". 4. The system automatically runs inspection on the captured snapshot. **Option B: Upload Image Files (all hardware types)** 1. In the sidebar, locate the Upload Images (JPG/PNG) section. 2. Click Browse files, or drag-and-drop image files directly onto the upload area. 3. Select one or more .jpg, .jpeg, or .png files and click Open. 4. A progress bar will appear in the main area while the AI processes each image. 5. Do NOT close or refresh the browser tab during processing - this will cancel the inspection. 

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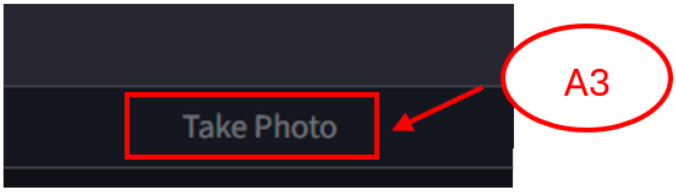
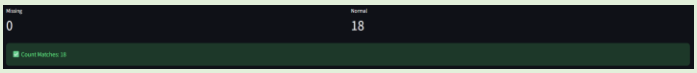
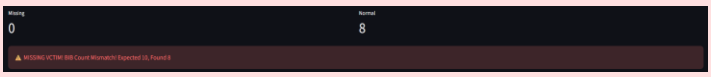
Step / Description	Action
	
Step 4 *Review Inspection Results*	After processing, results appear in the main content area as one expandable panel per image. Each panel contains: • Original image - the raw uploaded photo. • Result image - the AI-annotated photo with colored bounding boxes. **Reading the bounding boxes:**

Box Colour	Meaning
RED	missing_vctim - VCTIM component is absent at this location. Action required.
BLUE	normal - VCTIM component is present and detected correctly.

- Each box label includes a confidence value (ex: missing_vctim 0.87). Higher values indicate greater AI certainty.
- Below the images, numeric metrics show: Missing count and Normal count for that image.

Step 5
Interpret Validation Alert

Directly beneath the metrics, the system shows an automatic validation check comparing total detections vs. Expected DUT Amount:

Alert	Display	Action Required
PASS	Count Matches: N 	No action needed. Counts are correct.
MISMATCH	MISSING VCTIM! DUT Count Mismatch! Expected X, Found Y 	Investigate the unit. See troubleshooting section.

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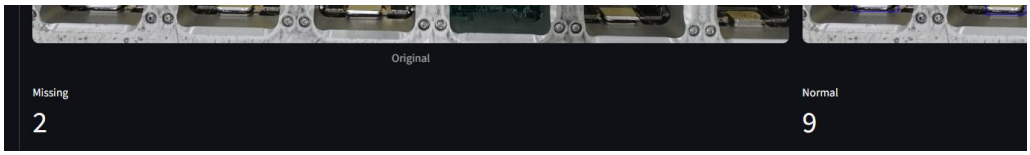
Step / Description	Action
	IMPORTANT: A MISMATCH alert means either: (a) a VCTIM is genuinely absent, (b) the Expected Socket Amount is set incorrectly, or (c) image quality caused a missed detection. Even if no RED boxes appear, a mismatch alert indicates the AI may have entirely missed a socket location. Manual visual inspection is required.
Step 6 *Annotate the Report (Optional)*	Below each image result, the Report Annotations section allows you to attach identifiers and notes to the inspection record. **Unit ID (Station Scanner or Manual Entry)** 1. Locate the Unit ID field below the result images. 2. To use the **barcode scanner**: click the Scan button next to the Unit ID field, then scan the barcode label on the unit. The scanned value populates the field automatically. 3. To **type manually**: click the Unit ID text box and type the identifier (ex: UNIT-001). **Comments / Notes** 4. Click the Comments text box and type any relevant notes. *Examples: False positive - component visually confirmed present, Re-inspection required, Image partially obscured.*
Step 7 *Review Batch Summary*	1. After all images in the batch have been processed, scroll to the bottom of the page. 2. The Batch Summary section shows the total defect count (sum of all Missing detections) and total passed count (sum of all Normal detections) across all uploaded images. 3. Use this as a quick pass/fail dashboard for the entire inspection batch before generating the report. 
Step 8	1. Scroll to the Generate Report section at the bottom of the page.

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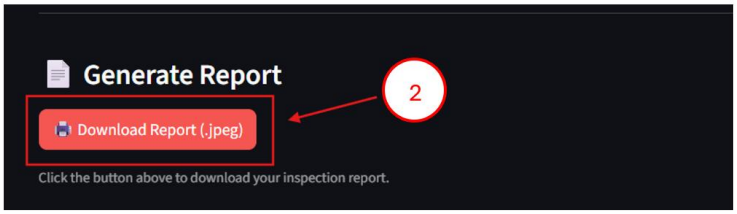
Step / Description	Action
Download Inspection Report	2. Click the Download Report (.jpeg) button.  3. The JPEG report file is saved to your browser's default download folder. *The report filename follows the pattern: <UnitID>_YYYYMMDD_HHMMSS.jpg (or inspection_report_VCTIM_Detection_<timestamp>.jpg if no Unit ID was entered).* 4. The report contains: header (mode, device, timestamp), batch summary table, side-by-side original vs. result images, per-image metrics, validation alerts, and any Unit ID / comments entered.

Analysis Outcomes - What to Expect

#	Outcome	What the System Shows	Recommended Action
1	All VCTIMs Present - No Anomaly	All boxes are BLUE. Validation alert is GREEN (Count Matches). Missing = 0.	No action required. Annotate Unit ID if needed and download report.
2	Missing VCTIM Detected	One or more RED (missing_vctim) boxes appear. Missing count > 0.	Flag the unit. Do not proceed to next step of production. Escalate per site procedure.
3	DUT Count Mismatch (no RED boxes but count is wrong)	No RED boxes, but MISMATCH alert appears (total found ≠ Expected DUT Amount).	Verify Expected DUT Amount setting. If correct, perform manual visual inspection. Check image quality (lighting, focus, framing).

6. Pass / Fail Criteria

8.0 PASS Criteria	8.0 FAIL Criteria
All of the following must be true: • Validation alert shows green Count Matches. • Missing VCTIM count = 0 for all images in the batch. • Total detected VCTIMs equals the Expected DUT Amount per image.	**Any of the following conditions:** • One or more missing_vctim detections (RED box) in any image. • DUT Count Mismatch alert (red) on any image - even with zero RED boxes.

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Troubleshooting

Issue / Symptom	Resolution
App does not load at localhost:8501	Verify Docker container is running (docker-compose up). Check your network connection. Contact your system administrator.
GPU selected but warning shows 'CUDA not available'	No compatible NVIDIA GPU detected, or CUDA drivers are not installed. The system automatically uses CPU. Contact IT if GPU acceleration is expected.
Mismatch alert but board appears correct	1) Verify Expected DUT Amount matches the product spec. 2) Check image is well-lit, in-focus, and all sockets are fully in-frame. 3) Try lowering the Detection Confidence slider slightly (ex: 0.25 → 0.20).
False positive - RED box on a present component	1) Increase Detection Confidence (ex: 0.25 → 0.35). 2) Improve lighting and recapture. 3) Note as 'False positive, visually confirmed' in the Comments field.
Progress bar stalls / app crashes	Do not refresh. Wait up to 60 seconds. If no change, refresh the page and re-upload images. Large batches (50+ images) on CPU may take several minutes.
Results appear stale after changing settings	Results are cached per file set + configuration. Changing the Detection Confidence or re-uploading files forces a fresh re-inspection.

8. Revision History

Revision	Date	Description	Owner
0	2026	Initial release - VCTIM Inspection BKM	
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